

Verifying the Future: Lean's Role in Building Reliable Systems

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How can we ensure that our most critical software and hardware systems behave exactly as intended, and that every proof of correctness is independently verifiable?

Before Lean, there was Z3

Z3 is a state-of-the-art SMT solver.

Z3 powered **bug-finding** pipelines like Microsoft's SAGE fuzzing tool: caught thousands of defects.

But for whole-program verification, proofs were brittle: minor code edits broke solver traces.

The Lean project, started in 2013, aimed at merging interactive and automated theorem proving.



Lean is an open-source programming language and proof assistant that is transforming how we approach mathematics, software verification, and AI.

Lean provides **machine-checkable proofs**. It addresses the “trust bottleneck”.

Lean opens up new possibilities for collaboration.

Lean and its tooling are implemented in Lean, and is very **extensible**.

Rapid self-improvement (flywheel).

Lean is based on dependent type theory

An example:

```
structure IndexMap (α : Type u) (β : Type v) [BEq α] [Hashable α] where
  private indices : HashMap α Nat
  private keys : Array α
  private values : Array β
  private size_keys' : keys.size = values.size := by grind
  private WF : ∀ (i : Nat) (a : α), keys[i]? = some a ↔ indices[a]? = some i := by grind
```

Full example [here](#).

An example:

```
structure IndexMap (α : Type u) (β : Type v) [BEq α] [Hashable α] where
  private indices : HashMap α Nat
  private keys : Array α
  private values : Array β
  private size_keys' : keys.size = values.size := by grind
  private WF : ∀ (i : Nat) (a : α), keys[i]? = some a ↔ indices[a]? = some i := by grind
```

```
def insert [LawfulBEq α] (m : IndexMap α β) (a : α) (b : β) : IndexMap α β :=
  match h : m.indices[a]? with
  | some i =>
    { indices := m.indices
      keys := m.keys.set i a
      values := m.values.set i b }
  | none =>
    { indices := m.indices.insert a m.size
      keys := m.keys.push a
      values := m.values.push b }
```

An example:

```
/-! ### Verification theorems -/

attribute [local grind] getIdx findIdx insert

@[grind] theorem getIdx_findIdx (m : IndexMap  $\alpha$   $\beta$ ) (a :  $\alpha$ ) (h : a  $\in$  m) :
  m.getIdx (m.findIdx a h) = m[a] := by grind

@[grind] theorem mem_insert (m : IndexMap  $\alpha$   $\beta$ ) (a a' :  $\alpha$ ) (b :  $\beta$ ) :
  a'  $\in$  m.insert a b  $\Leftrightarrow$  a' = a  $\vee$  a'  $\in$  m := by
  grind

@[grind] theorem getElem_insert (m : IndexMap  $\alpha$   $\beta$ ) (a a' :  $\alpha$ ) (b :  $\beta$ ) (h : a'  $\in$  m.insert a b) :
  (m.insert a b)[a']'h = if h' : a' == a then b else m[a'] := by
  grind

@[grind] theorem findIdx_insert_self (m : IndexMap  $\alpha$   $\beta$ ) (a :  $\alpha$ ) (b :  $\beta$ ) :
  (m.insert a b).findIdx a (by grind) = if h : a  $\in$  m then m.findIdx a h else m.size := by
  grind
```



Lean is an IDE for formal verification

Lean is a development environment for formal verification

The math community using Lean is growing rapidly. They love the system.

Lean is used in several software verification projects at AWS since 2023.

Lean has a rich user interface.

Theorem proving in Lean is an interactive game

The screenshot displays the Lean IDE interface. On the left, the source code for a theorem proof is shown:

```
Odd.lean > square_of_odd_is_odd
1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    done
```

On the right, the 'Lean Infoview' pane shows the current goal state:

```
▼ Odd.lean:7:2
▼ Tactic state
1 goal
  n : ℕ
  ⊢ odd n → odd (n * n)
```

A blue box labeled "The 'game board'" points to the goal state in the Infoview. The Infoview also shows 'Messages (1)' and 'All Messages (2)'.

"You have written my favorite computer game", Kevin Buzzard

Theorem proving in Lean is an interactive game

Odd.lean > `square_of_odd_is_odd`

```
1 import Mathlib
2
3 def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5 -- Prove that the square of an odd number is always odd
6 theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7   intro ⟨k1, e1⟩
8   done
9
10
11
12
```

A "game move", aka "tactic"

Lean Infoview

▼ Odd.lean:8:2

▼ Tactic state

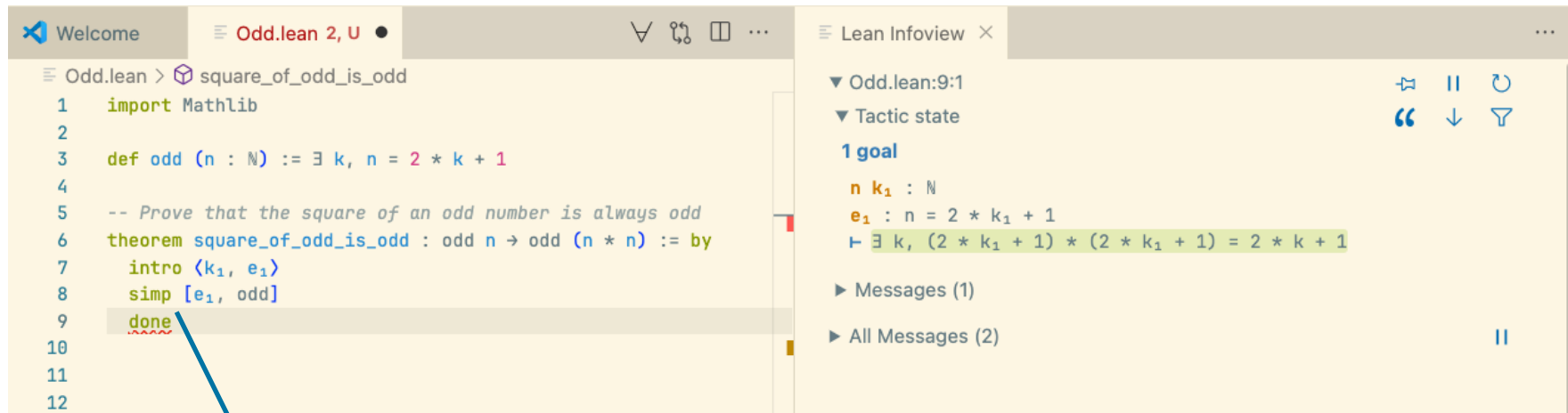
1 goal

$n \ k_1 : \mathbb{N}$
 $e_1 : n = 2 * k_1 + 1$
└ odd (n * n)

► Messages (1)

► All Messages (2)

Theorem proving in Lean is an interactive game



```
Odd.lean > square_of_odd_is_odd
1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    intro (k₁, e₁)
8    simp [e₁, odd]
9    done
10
11
12
```

Lean Infoview

▼ Odd.lean:9:1

▼ Tactic state

1 goal

n k₁ : ℕ

e₁ : n = 2 * k₁ + 1

└ ∃ k, (2 * k₁ + 1) * (2 * k₁ + 1) = 2 * k + 1

► Messages (1)

► All Messages (2)

The “game move” `simp`, the simplifier, is one of the most popular moves in our game

Theorem proving in Lean is an interactive game

The screenshot displays the Lean IDE interface. The left pane shows the source code for a file named `Odd.lean`. The code defines an `odd` predicate and a theorem `square_of_odd_is_odd`. The proof of the theorem is in progress, with the `use` tactic being applied to the goal. A blue arrow points from the `use` tactic in the code to a callout box. The right pane shows the 'Lean Infoview' for the current goal, displaying the tactic state and the current goal.

```
Odd.lean > square_of_odd_is_odd
1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    intro (k₁, e₁)
8    simp [e₁, odd]
9    use 2 * k₁ * k₁ + 2 * k₁
10   done
11
12
```

Lean Infoview

▼ Odd.lean:10:1

▼ Tactic state

1 goal

▼ case h

n k₁ : ℕ

e₁ : n = 2 * k₁ + 1

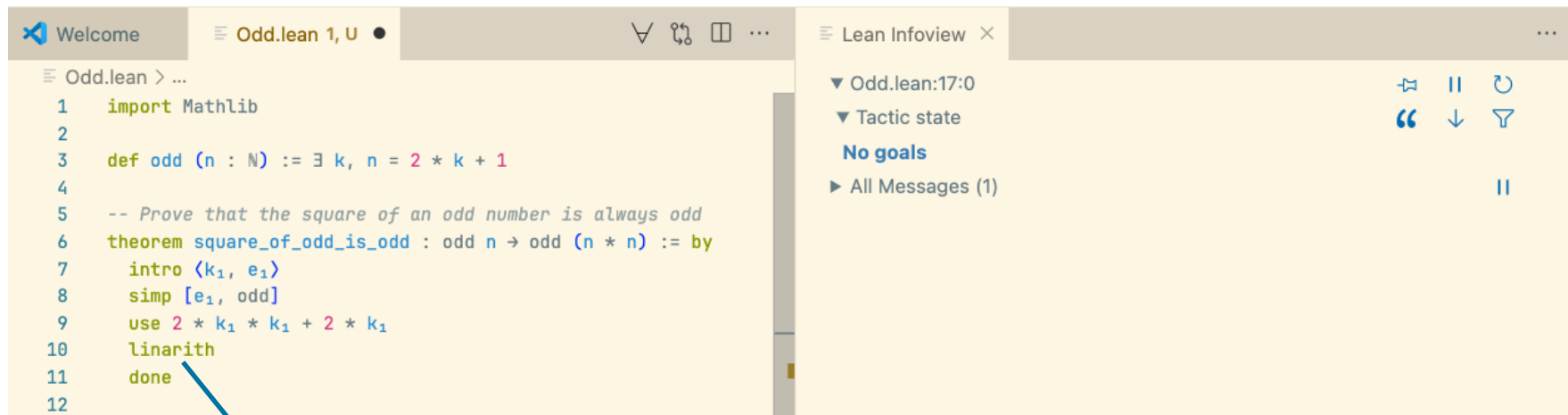
├ (2 * k₁ + 1) * (2 * k₁ + 1) = 2 * (2 * k₁ * k₁ + 2 * k₁) + 1

► Messages (1)

► All Messages (2)

The “game move” `use` is the standard way of proving statements about existentials

Theorem proving in Lean is an interactive game



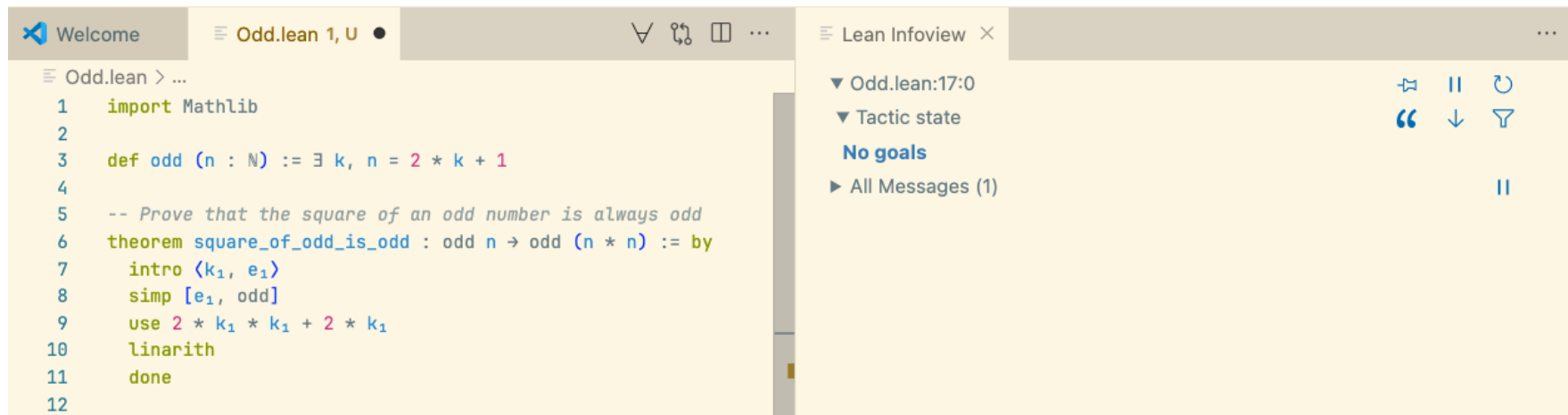
```
1 import Mathlib
2
3 def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5 -- Prove that the square of an odd number is always odd
6 theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7   intro (k₁, e₁)
8   simp [e₁, odd]
9   use 2 * k₁ * k₁ + 2 * k₁
10  linarith
11  done
12
```

Lean Infoview

- Odd.lean:17:0
- Tactic state
- No goals
- All Messages (1)

We complete this level using `linarith`, the linear arithmetic, move

Theorem proving in Lean is an interactive **and addictive** game



The screenshot shows the Lean IDE interface. The left pane displays the source code for a file named `Odd.lean`. The code defines an odd number and proves that its square is also odd. The right pane shows the 'Lean Infoview' for the current tactic state, indicating 'No goals' and 'All Messages (1)'. The interface includes a top bar with file tabs and a command palette, and a bottom status bar.

```
1 import Mathlib
2
3 def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5 -- Prove that the square of an odd number is always odd
6 theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7   intro ⟨k₁, e₁⟩
8   simp [e₁, odd]
9   use 2 * k₁ * k₁ + 2 * k₁
10  linarith
11  done
12
```

Lean Infoview

- ▼ Odd.lean:17:0
- ▼ Tactic state
- No goals
- All Messages (1)

"You can do 14 hours a day in it and not get tired and feel kind of high the whole day. You're constantly getting positive reinforcement", Amelia Livingston



Mathlib

The Lean Mathematical Library supports a wide range of projects.

It is an open-source **collaborative project** with over 500 contributors and 1.8M LoC.

"I'm investing time now so that somebody in the future can have that amazing experience",

Heather Macbeth



Quanta magazine

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[Mathematics](#)

[Biology](#)

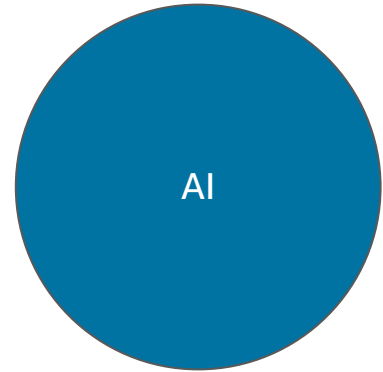
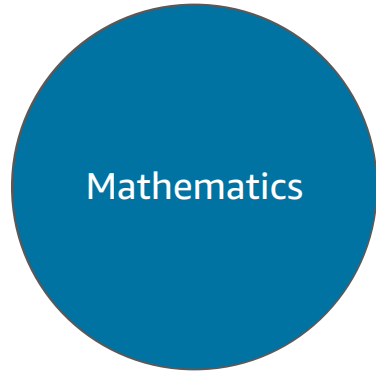
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FOUNDATIONS OF MATHEMATICS

Building the Mathematical Library of the Future



Mathematics

Preamble: the Perfectoid Spaces Project

Kevin Buzzard, Patrick Massot, Johan Commelin

Goal: Demonstrate that we can **define complex mathematical objects** in Lean.

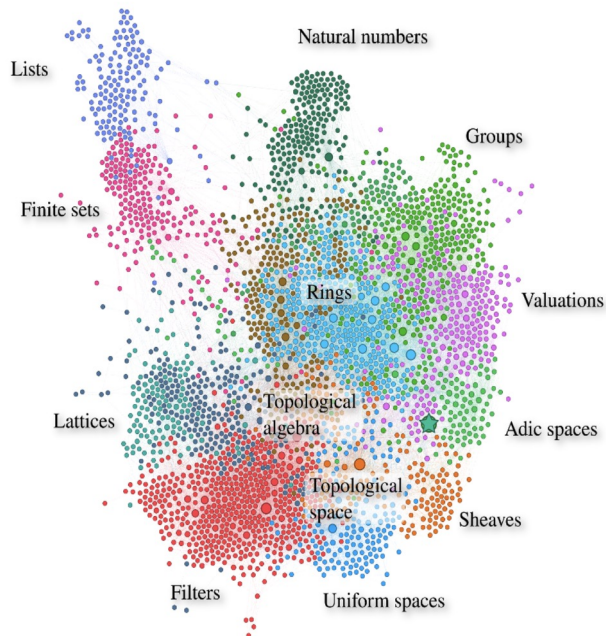
They translated Peter Scholze's definition into a form a computer can understand.

It not only achieved its goals but also demonstrated to the math community that **formal objects can be visualized and inspected with computer assistance**.

Math is now **data** that can be **processed, transformed**, and **inspected** in various ways.

Preamble: the Perfectoid Spaces Project (cont.)

Kevin Buzzard, Patrick Massot, Johan Commelin



mathoverflow

Home

What are "perfectoid spaces"?

▲

Here is a completely different kind of answer to this question.

72

A perfectoid space is a term of type `PerfectoidSpace` in the [Lean theorem prover](#).

▼

Here's a quote from the source code:

```

structure perfectoid_ring (R : Type) [Huber_ring R] extends Tate_ring R : Prop :=
  (complete : is_complete_hausdorff R)
  (uniform : is_uniform R)
  (ramified : ∃ w : pseudo_uniformizer R, w^p ∣ p in R)
  (Frobenius : surjective (Frob R/p))

```



Mathlib > RingTheory > Finiteness.lean

```
355
356 theorem F6.stabilizes_of_iSup_eq {M' : Submodule R M} (hM' : M'.F6) (N : ℕ → Submodule R M)
357   (H : iSup N = M') : ∃ n, M' = N n := by
358   obtain ⟨S, hS⟩ := hM'
359   have : ∀ s : S, ∃ n, (s : M) ∈ N n := fun s =>
360     (Submodule.mem_iSup_of_chain N s).mp
361     (by
362       rw [H, ← hS]
363       exact Submodule.subset_span s.2)
364   choose f hf using this
365   use S.attach.sup f
366   apply le_antisymm
367   · conv_lhs => rw [← hS]
368     rw [Submodule.span_le]
369     intro s hs
370     exact N.2 (Finset.le_sup <| S.mem_attach ⟨s, hs⟩) (hf _)
371   · rw [← H]
372     exact le_iSup _ _
```

▼ Finiteness.lean:365:2

▼ Tactic state

1 goal

▼ case intro

R : Type u_1

M : Type u_2

inst² : Semiring R

inst¹ : AddCommMonoid M

inst : Module R M

M' : Submodule R M

N : ℕ → Submodule R M

H : iSup ↑N = M'

S : Finset M

hS : span R ↑S = M'

f : { x // x ∈ S } → ℕ

hf : ∀ (s : { x // x ∈ S }), ↑s ∈ N (f s)

⊢ ∃ n, M' = N n

Mathlib > RingTheory > Finiteness.lean

```

355
356 theorem FG.stabilizes_of_iSup_eq {M' : Submodule R M} (hM' : M'.FG) (N : ℕ → Submodule R M)
357   (H : iSup N = M') : ∃ n, M' = N n := by
358   obtain ⟨S, hS⟩ := hM'
359   have : ∀ s : S, ∃ n, (s : M) ∈ N n := fun s =>
360     (Submodule.mem_iSup_of_chain N s).mp
361     (by
362       rw [H, ← hS]
363       exact Submodule.subset_span s.2)
364   choose f hf using this
365   use S.attach.sup f
366   apply le_antisymm
367   · conv_lhs => rw [← hS]
368     rw [Submodule.span_le]
369     intro s hs
370     exact N.2 (Finset.le_sup <| S.mem_attach ⟨s, hs⟩) (hf _)
371   · rw [← H]
372     exact le_iSup _ _
---
```

▼ Finiteness.lean:365:2

▼ Tactic state

1 goal

▼ case intro

R : Type u_1

M : Type u_2

inst² : Semiring R

inst¹ : AddCommMonoid M

inst : Module R M

M' : Submodule R M

N : ℕ → Submodule R M

H : iSup ↑N = M'

S : Finset M

hS : span R ↑S = M'

f : { x // x ∈ S } → ℕ

hf : ∀ (s : { x // x ∈ S }), ↑s ∈ N (f s)

⊢ ∃ n, M' = N n



Mathlib > RingTheory > Finiteness.lean

355

356 **theorem** `FG.stabilizes_of_iSup_eq` `{M' : Submodule R M}` `(hM' : M'.FG)` `(N : ℕ → Submodule R M)`

Defs.lean ~/projects/mathlib4/Mathlib/Algebra/Module/Submodule - Definitions (1)

25 **assert_not_exists** `DivisionRing`

26

27 **open** `Function`

28

29 **universe** `u' u' u v w`

30

31 **variable** `{G : Type u''}` `{S : Type u'}` `{R : Type u}` `{M : Type v}` `{u :`

32

33 */-- A submodule of a module is one which is closed under vector oper*

34 *This is a sufficient condition for the subset of vectors in the su*

35 *to themselves form a module. -/*

36 **structure** `Submodule` `(R : Type u)` `(M : Type v)` `[Semiring R]` `[AddCommM`

37 `AddSubmonoid M, SubMulAction R M : Type v`

38

structure `Submodule` `(R : Type u)` `(`

▼ Finiteness.lean:356:44

▼ Expected type

`R` : Type `u_1`

`M` : Type `u_2`

`inst+4` : Semiring `R`

`inst+3` : AddCommMonoid `M`

`inst+2` : Module `R M`

`P` : Type `u_3`

`inst+1` : AddCommMonoid `P`

`inst+` : Module `R P`

`f` : `M →[R] P`

`⊢` Type `u_2`

► All Messages (0)



Mathlib > Algebra > Module > Submodule > Defs.lean > Submodule

```
34   This is a sufficient condition for the subset of vectors in the submodule
35   to themselves form a module. -/
36   structure Submodule (R : Type u) (M : Type v) [Semiring R] [AddCommMonoid M] [Module R M] extends
37     AddSubmonoid M, SubMulAction R M : Type v
```

Defs.lean ~/projects/mathlib4/Mathlib/Algebra/Group/Submonoid - Definitions (1)

```
84   add_decl_doc Submonoid.toSubsemigroup
85
86   /-- `SubmonoidClass S M` says `S` is a type of subsets `s ≤ M` that
87   and are closed under `(*)` -/
88   class SubmonoidClass (S : Type*) (M : outParam Type*) [MulOneClass M]
89     MulMemClass S M, OneMemClass S M : Prop
90
91   section
92
93   /-- An additive submonoid of an additive monoid `M` is a subset containing
94   closed under addition. -/
95   structure AddSubmonoid (M : Type*) [AddZeroClass M] extends AddSubsemigroup M
96     /-- An additive submonoid contains `0`. -/
97     zero_mem' : (0 : M) ∈ carrier
98
```

structure AddSubmonoid (M : Type*)

▼ Defs.lean:37:8

▼ Expected type

```
G : Type u''
S : Type u'
R† : Type u
M† : Type v
ι : Type w
R : Type u
M : Type v
inst†² : Semiring R
inst†¹ : AddCommMonoid M
inst† : Module R M
⊢ Type v
```

► All Messages (0)

The Challenge

In November of 2020, Peter Scholze posits the Liquid Tensor Experiment (LTE) challenge.

*"I spent much of 2019 **obsessed** with the proof of this theorem, **almost getting crazy over it**. In the end, we were able to get an argument pinned down on paper, but I think nobody else has dared to look at the details of this, and so I still have some small lingering doubts",*

Peter Scholze

The First Victory

Johan Commelin led a team with several members of the **Lean community** and announced the **formalization of the crucial intermediate lemma** that Scholze was unsure about, with only minor corrections, in **May 2021**.

"[T]his was precisely the kind of oversight I was worried about when I asked for the formal verification. [...] The proof walks a fine line, so if some argument needs constants that are quite a bit different from what I claimed, it might have collapsed", Peter Scholze

nature

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NEWS | 18 June 2021

**Mathematicians welcome
computer-assisted proof in 'grand
unification' theory**

Achieving the Unthinkable

The full challenge was completed in July 2022.

The team not only verified the proof but also simplified it.

Moreover, they did this without fully understanding the entire proof.

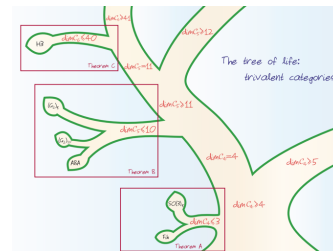
Johan, the project lead, reported that he could only see two steps ahead. **Lean was a guide.**

“The Lean Proof Assistant was really that: an assistant in navigating through the thick jungle that this proof is. Really, one key problem I had when I was trying to find this proof was that I was essentially unable to keep all the objects in my RAM, and I think the same problem occurs when trying to read the proof”, Peter Scholze

Here is a concrete example from quantum algebra. It comes from a classification result involving quantum $SO(3)$ categories. Specifically, the condition that certain relations among trivalent graphs imply a constraint on the parameters d , t , and c :

```
example {α} [CommRing α] [IsCharP α 0] (d t c : α) (d_inv PS03_inv : α)
  (Δ40 : d^2 * (d + t - d * t - 2) *
    (d + t + d * t) = 0)
  (Δ41 : -d^4 * (d + t - d * t - 2) *
    (2 * d + 2 * d * t - 4 * d * t^2 + 2 * d * t^4 + 2 * d^2 * t^4 - c * (d + t + d * t))) = 0)
  (_ : d * d_inv = 1)
  (_ : (d + t - d * t - 2) * PS03_inv = 1) :
  t^2 = t + 1 := by grind
```

From: "Categories generated by a trivalent vertex", Morrison, Peters, and Snyder



Automating Quantum Algebra

```
example {α} [CommRing α] [IsCharP α 0] (d t c : α) (d_inv PS03_inv : α)
  (Δ40 : d^2 * (d + t - d * t - 2) *
    (d + t + d * t) = 0)
  (Δ41 : -d^4 * (d + t - d * t - 2) *
    (2 * d + 2 * d * t - 4 * d * t^2 + 2 * d * t^4 + 2 * d^2 * t^4 - c * (d + t + d * t))) = 0)
  (_ : d * d_inv = 1)
  (_ : (d + t - d * t - 2) * PS03_inv = 1) :
  t^2 = t + 1 := by grind
```

This is not a toy: it encodes a real algebraic constraint derived from relations among diagrams in a pivotal tensor category.

Lean can handle this kind of reasoning automatically, in [milliseconds](#).



Automating Quantum Algebra

We can explore new mathematical and physical structures, from topological quantum fields theories to fusion categories.

Lean is helping researchers reason reliably about complex symbolic systems that were previously handled only by hand or with unverified computer algebra.

`grind` **is just another move in our interactive game.**

Should we trust Lean?

Lean has a small trusted proof checker.

Do I need to trust the checker?

No, **you can export your proof**, and use external checkers. There are checkers implemented in C/C++, Rust, Lean, etc.

You can implement your own checker.



What did we learn?

Machine-checkable proofs enable a new level of **collaboration** in mathematics.

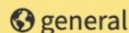
The power of the **community**.

We don't need to trust our automation/moves.

It is not just about proving but also understanding complex objects and proofs, getting new insights, and navigating through the “thick jungles” that are **beyond our cognitive abilities**.

What did we learn?

Another unexpected benefit of formal mathematics: **auto refactoring** and **generalization**.



general

An example of why formalization is useful

Mar 31



Riccardo Brasca EDITED

7:53 AM

I really like what is going on with [#12777](#). [@Sebastian Monnet](#) proved that if E , F and K are fields such that `finite_dimensional F E`, then `fintype (E →a [F] K)`. We already have [docs#field.alg_hom.fintype](#), that is exactly the same statement with the additional assumption `is_separable F E`.

The interesting part of the PR is that, with the new theorem, the linter will automatically flag all the theorem that can be generalized (for free!), removing the separability assumption. I think in normal math this is very difficult to achieve, if I generalize a 50 years old paper that assumes `p ≠ 2` to all primes, there is no way I can manually check and maybe generalize all the papers that use the old one.



3



5

Software

Lean in Software Verification

Lean is a programming language, and is used in **many software verification projects**.

You can write code and reason about it simultaneously.

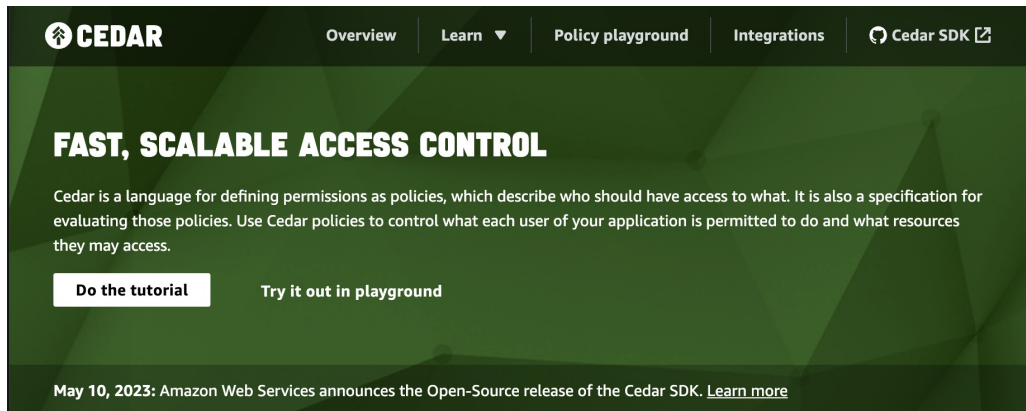
You can prove that your code has the properties you expect.

“Testing can show the presence of bugs, but not their absence”, E. Dijkstra



Cedar

<https://www.cedarpolicy.com/>

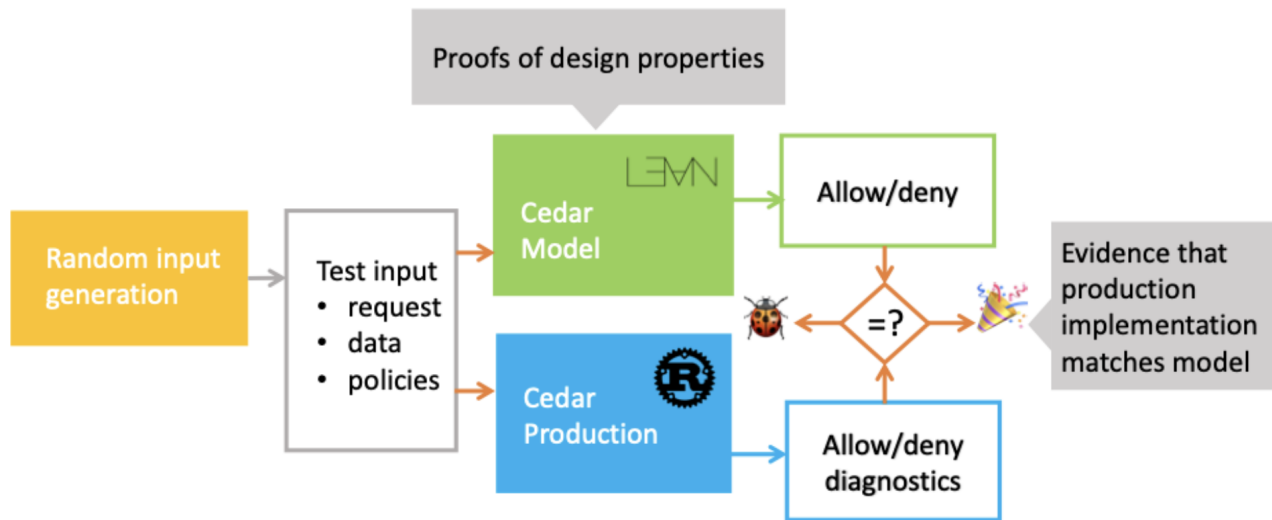


The screenshot shows the Cedar website homepage. At the top is a dark navigation bar with the Cedar logo on the left and links for Overview, Learn (with a dropdown arrow), Policy playground, Integrations, and Cedar SDK (with an external link icon). The main content area has a dark green background with a geometric pattern. It features the heading "FAST, SCALABLE ACCESS CONTROL" in white. Below this is a paragraph explaining that Cedar is a language for defining permissions as policies. Two buttons are present: "Do the tutorial" (highlighted with a white border) and "Try it out in playground". At the bottom, a dark banner contains the text "May 10, 2023: Amazon Web Services announces the Open-Source release of the Cedar SDK. [Learn more](#)".

<https://github.com/cedar-policy/cedar-spec>

```
def isAuthorized (req : Request) (entities : Entities) (policies : Policies) : Response :=
  let forbids := satisfiedPolicies .forbid policies req entities
  let permits := satisfiedPolicies .permit policies req entities
  let erroringPolicies := errorPolicies policies req entities
  if forbids.isEmpty && !permits.isEmpty
  then { decision := .allow, determiningPolicies := permits, erroringPolicies }
  else { decision := .deny, determiningPolicies := forbids, erroringPolicies }
```

Cedar



Takeaway: “We’ve found Lean to be a great tool for verified software development. You get a full-featured programming language, fast proof checker and runtime, and a familiar way to build both models and proofs”



Cedar

To learn more about Cedar:

<https://aws.amazon.com/blogs/opensource/lean-into-verified-software-development/>



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Lean Into Verified Software Development

by Kesha Hietala and Emina Torlak | on 08 APR 2024 | in [Amazon Verified Permissions](#), [Open Source](#), [Security](#), [Identity](#), & [Compliance](#), [Technical How-to](#) | [Permalink](#) | [Comments](#) | [Share](#)

Resources

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Differential Privacy

A mathematical framework that ensures the **privacy of individuals** in a dataset by adding controlled **random noise** to the data.

Discrete sampling algorithms, like the **Discrete Gaussian Sampler**, are used to add carefully calibrated noise to data.

What may go wrong if a buggy sampler is used?

Privacy Violations: leakage of sensitive information

Incorrect Results: distorted analysis results

SampCert

A project led by **Jean-Baptiste Tristan** at AWS.

An **open-source** Lean library of formally **verified differential privacy primitives**.

Tristan's implementation is not only verified, but it is also **twice as fast as the previous one**.

He managed to implement **aggressive optimizations** because Lean served as a guide, ensuring that **no bugs** were introduced.

SampCert would not exist without Mathlib

SampCert is software, but its verification relies heavily on Mathlib.

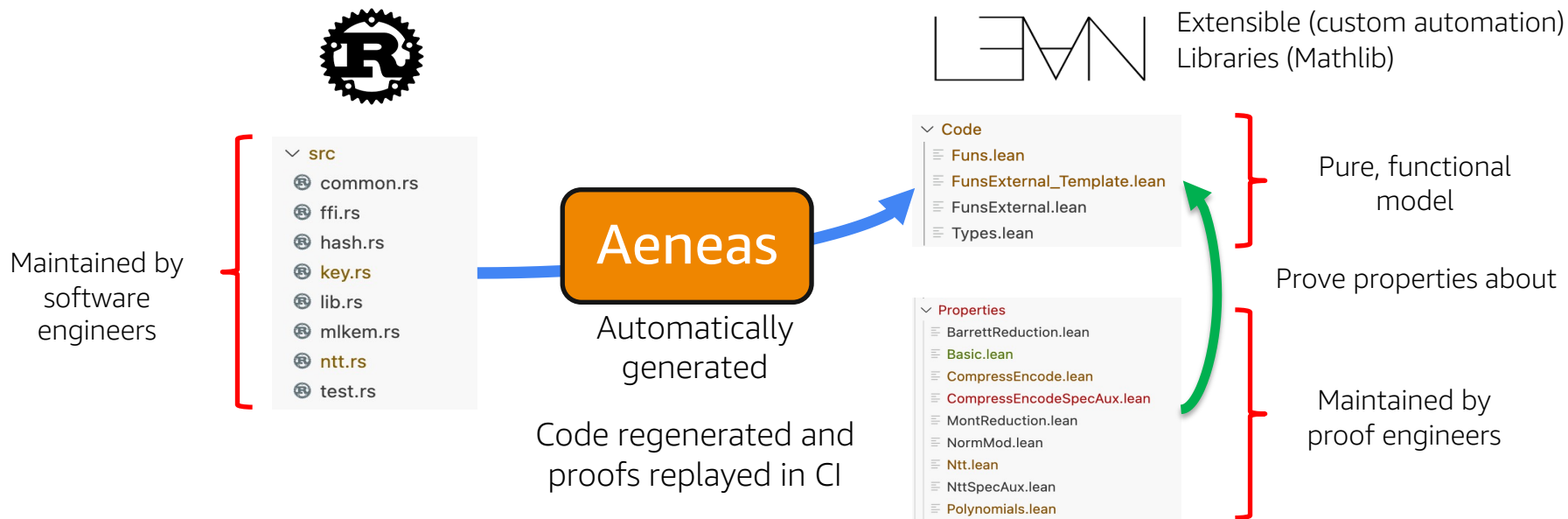
The verification of code addressing practical problems in data privacy depends on the formalization of mathematical concepts, from **Fourier analysis** to **number theory** and **topology**.

Verifying Cryptography with Aeneas at Microsoft

They verify (and fix/improve) the Rust code as written by software engineers.

Code is evolving (new optimizations for specific hardware): They must adapt to rewrites.

[Rewriting SymCrypt in Rust to modernize Microsoft's cryptographic library.](#)





KLR: a language and elaborators for machine learning kernels

Define a common representation for kernel functions with a precise formal semantics along with translations from common kernel languages to the KLR core language.

KLR is also open source.

```
private def evalTensorScalar (ts : TensorScalar) (t: ByteArray) : Err ByteArray := do
  match ts with
  | TensorScalar.mk op0 c0 rev0 op1 c1 rev1 =>
    let f0 <- evalAluOp op0
    let f1 <- evalAluOp op1
    let c0 := c0.toLEByteArray
    let c1 := c1.toLEByteArray
    apply2 f0 rev0 c0 f1 rev1 c1 t
```

KLR: a language and elaborators for machine learning kernels

KLR uses bit-vectors, fixed integers, etc.

```
private def decBV64 : DecodeM (BitVec 64) :=
  let u8_64 : DecodeM UInt64 := next >=> fun x => return x.toUInt64
  return ((<- u8_64) <<< 0 |||
    (<- u8_64) <<< 8 |||
    (<- u8_64) <<< 16 |||
    (<- u8_64) <<< 24 |||
    (<- u8_64) <<< 32 |||
    (<- u8_64) <<< 40 |||
    (<- u8_64) <<< 48 |||
    (<- u8_64) <<< 56).toBitVec
```

bv_decide: another powerful move

A verified bit-blaster by **Henrik Boving**, Josh Clune, Siddharth Bhat, and Alex Keizer

Uses LRAT proof producing SAT solvers: **Cadical**

```
/-  
Close a goal by:  
1. Turning it into a BitVec problem.  
2. Using bitblasting to turn that into a SAT problem.  
3. Running an external SAT solver on it and obtaining an LRAT proof from it.  
4. Verifying the LRAT proof using proof by reflection.  
-/  
syntax (name := bvDecideSyntax) "bv_decide" : tactic
```

“Blasting” popcount with bv_decide

```
def popcount : Stmt := imp {
  x := x - ((x >>> 1) &&& 0x55555555);
  x := (x &&& 0x33333333) + ((x >>> 2) &&& 0x33333333);
  x := (x + (x >>> 4)) &&& 0x0F0F0F0F;
  x := x + (x >>> 8);
  x := x + (x >>> 16);
  x := x &&& 0x0000003F;
}
```

```
def pop_spec (x : BitVec 32) : BitVec 32 :=
  go x 0 32
where
  go (x : BitVec 32) (pop : BitVec 32) (i : Nat) : BitVec 32 :=
    match i with
    | 0 => pop
    | i + 1 =>
      let pop := pop + (x &&& 1#32)
      go (x >>> 1#32) pop i
```

```
theorem popcount_correct :
```

```
  ∃ p, (run (Env.init x) popcount 8) = some p ∧ p "x" = pop_spec x := by
  simp [run, popcount, Expr.eval, Expr.BinOp.apply, Env.set, Value, pop_spec, pop_spec.go]
  bv_decide
```

“Blasting” popcount with bv_decide

Imp.lean > {} Imp.Stmt > popcount_correct

```
50 theorem popcount_correct :
51   ∃ p, (run (Env.init x) popcount 8) = some p
52   simp [run, popcount, Expr.eval, Expr.BinOp.app
53   bv_decide
54
```

▼Tactic state

1 goal

```
x : Value
├ ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) + ((x - (x >> 1 &&&
1431655765#32)) >> 2 &&& 858993459#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32)) >>
      4 &&&
        252645135#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32)) >>
      4 &&&
        252645135#32) >>
      8 +
  (((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32)) >>
      4 &&&
        252645135#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32) +
  ((x - (x >> 1 &&& 1431655765#32) &&& 858993459#32) +
    ((x - (x >> 1 &&& 1431655765#32)) >> 2 &&& 858993459#32)) >>
      4 &&&
        252645135#32) >>
      8) >>
    16 &&&
      63#32 =
  (x &&& 1#32) + (x >> 1 &&& 1#32) + (x >> 2 &&& 1#32) + (x >> 3 &&& 1#32) + (x >>
4 &&& 1#32) +
```

grind (again)

```
example (x : BitVec 16) : (x + 256)*(x - 256) = x^2 := by
  grind
```

```
def siftDown (a : Array Int) (root : Nat) (e : Nat) (h : e ≤ a.size := by grind) : Array Int :=
  if _ : leftChild root < e then
    let child := leftChild root
    let child := if _ : child+1 < e then
      if a[child] < a[child + 1] then child + 1 else child
    else child
    if a[root] < a[child] then
      let a := a.swap root child
      siftDown a child e
    else a
  else a
termination_by e - root

theorem siftDown_size {a root e h} : (siftDown a root e h).size = a.size := by
  fun_induction siftDown <=> grind [siftDown]
```

grind diagnostics

```
example {α} (as bs cs : Array α) (v₁ v₂ : α)
  (i₁ i₂ j : Nat)
  (h₁ : i₁ < as.size)
  (h₂ : bs = as.set i₁ v₁)
  (h₃ : i₂ < bs.size)
  (h₄ : cs = bs.set i₂ v₂)
  (h₅ : i₁ ≠ j)
  (h₆ : j < cs.size)
  (h₆ : j < as.size)
  : cs[j] = as[j] := by
```

grind

```
`grind` failed
▼case grind
α : Type u_1
as bs cs : Array α
v₁ v₂ : α
i₁ i₂ j : Nat
h₁ : i₁ + 1 ≤ as.size
h₂ : bs = as.set i₁ v₁ ...
h₃ : i₂ + 1 ≤ bs.size
h₄_1 : cs = bs.set i₂ v₂ ...
h₄ : ¬i₁ = j
h₅ : j + 1 ≤ cs.size
h₆ : j + 1 ≤ as.size
h : ¬cs[j] = as[j]
└ False
```

```
[grind] Goal diagnostics ▼
[facts] Asserted facts ►
[eqc] True propositions ►
[eqc] False propositions ►
[eqc] Equivalence classes ►
[ematch] E-matching patterns ►
[cutsat] Assignment satisfying linear constraints ▼
  [assign] i₁ := 0
  [assign] i₂ := 1
  [assign] j := 1
  [assign] as.size := 2
  [assign] bs.size := 2
  [assign] cs.size := 2
```

Community driven proof automation

The Lean community is also actively developing automation.

[Lean-SMT: An SMT tactic for discharging proof goals in Lean](#)

[LeanHammer](#): an automated reasoning tool for Lean which brings together multiple proof search and reconstruction techniques and combine them into one tool.



What did we learn?

Machine-checkable proofs enable you to **code without fear**.

Powerful proof automation.

Industrial projects: Verified compilers, policy languages, cryptographic libraries, etc.

Many more at the **Lean Project Registry**: <https://reservoir.lean-lang.org/>

 | science

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AUTOMATED REASONING

How the Lean language
brings math to coding
and coding to math

AI



Lean Enables **Verified** AI for Mathematics and Code

LLMs are powerful tools, but they are prone to **hallucinations**.

In Math, a **small mistake can invalidate the whole proof**.

Imagine manually checking an AI-generated proof with the size and complexity of FLT.

The informal proof is **over 200 pages**.

Buzzard estimates a formal proof will require more than **1M LoC** on top of Mathlib.

Machine-checkable proofs are the antidote to hallucinations.

Synthetic Data Generation

LLMs require **vast amounts of data** for training.

Lean mathematical libraries provide valuable, **correct-by-construction training data**.

Tools like [lean-training-data](#), by **Kim Morrison**, extract data that includes the “game board” before and after each “move”.

The screenshot displays the Lean IDE interface. The main editor shows a Lean script for proving that the square of an odd number is odd. The script includes an import for Mathlib, a definition for odd numbers, and a theorem statement. The proof is completed using the 'done' tactic. The right-hand side of the interface shows the Lean Infoview, which displays the current goal and the tactic state. The goal is to prove that for any odd number n , n^2 is odd. The tactic state shows the current goal and the tactics used to reach it.

```
Odd.lean > square_of_odd_is_odd
1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    intro ⟨k₁, e₁⟩
8    simp [e₁, odd]
9    done
10
11
12
```

Lean Infoview

Odd.lean:9:1

Tactic state

1 goal

$n \ k_1 : \mathbb{N}$

$e_1 : n = 2 * k_1 + 1$

$\vdash \exists k, (2 * k_1 + 1) * (2 * k_1 + 1) = 2 * k + 1$

Messages (1)

All Messages (2)

Synthetic Data Generation

LLMs require **vast amounts of data** for training.

Lean mathematical libraries provide valuable, **correct-by-construction training data**.

Tools like [lean-training-data](#), by **Kim Morrison**, extract data that includes the “game board” before and after each “move”.

We can search for alternative proofs and automatically verify their correctness.

AlLean, a project led by **Soonho Kong** at AWS, uses Lean to generate **new synthetic theorems** that are correct by construction.

AI Proof Assistants

Several groups are developing AI that suggests the **next move**(s) in Lean's interactive proof game.

[LeanDojo](#) is an open-source project from Caltech, and everything (model, datasets, code) is open.

[OpenAI](#) and [Meta AI](#) have also developed AI assistants for Lean.

Claude 4 is fantastic on Lean code. Their [System Card](#) contains a Lean example.



The New York Times

Artificial Intelligence > A.I.'s Math Problem A.I. Training Data Disappears Microsoft's Risk-Taker Fine Print Changes Quiz: Fake or Real Images?

Move Over, Mathematicians, Here Comes AlphaProof

A.I. is getting good at math — and might soon make a worthy collaborator for humans.

Share full article



Ring the gong at Google Deepmind's London headquarters, a ritual to celebrate each A.I. milestone, including its recent triumph of reasoning at the International Mathematical Olympiad. Google Deepmind

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main

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Rekile and Paul-Lez Fix: AMS codes (#185) ed8a809 · yesterday

.devcontainer	feat: Add gitpod integration (#181)	2 days ago
.github	Fix caching issues with the doc buil...	last week
.vscode	vscode settings (#164)	5 days ago
FormalConjectures	Fix: AMS codes (#185)	yesterday
docbuild	Fix caching issues with the doc buil...	last week
scripts	ci: add a copyright header check (#...	2 weeks ago
.gitignore	move OpenProblems to third_party	2 months ago
.gitpod.yml	feat: Add gitpod integration (#181)	2 days ago
.mailmap	chore: add .mailmap (#60)	2 weeks ago

A collection of formalized statements of conjectures in Lean.

google-deepmind.github.io/fo...

formal-mathematics lean4

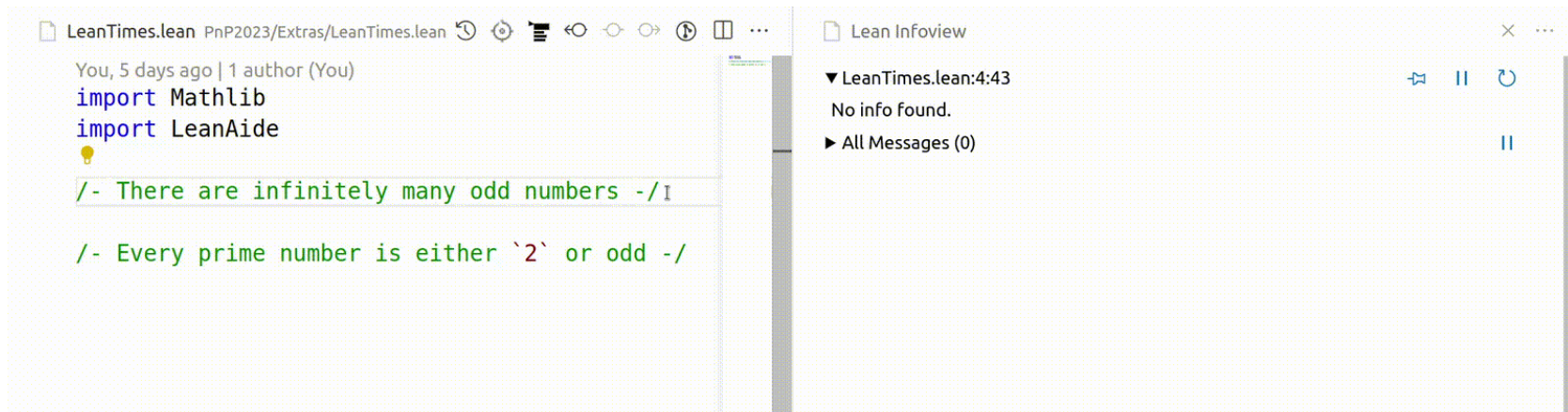
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Auto-formalization

The process of converting natural language into a formal language like Lean.

It is much **easier to learn to read Lean than to write it**.

[LeanAide](#) is one of the auto-formalization tools available for Lean.



```
LeanTimes.lean PnP2023/Extras/LeanTimes.lean
You, 5 days ago | 1 author (You)
import Mathlib
import LeanAide
💡
/- There are infinitely many odd numbers -/
/- Every prime number is either `2` or odd -/

Lean Infoview
▼ LeanTimes.lean:4:43
No info found.
► All Messages (0)
```

What did we learn?

Machine-checkable proofs enable **AI that does not hallucinate**.

LLMs are getting better and better at explaining Lean code.

In an era of big data and LLMs, machine-checkable proofs ensure trust in results.

AI systems that prove rather than guess.

Before we wrap up...



Lean Enables Decentralized Collaboration

Lean is Extensible

Users extend Lean using Lean itself.

Lean is implemented in Lean.

You can make it your own.

You can create your own moves.

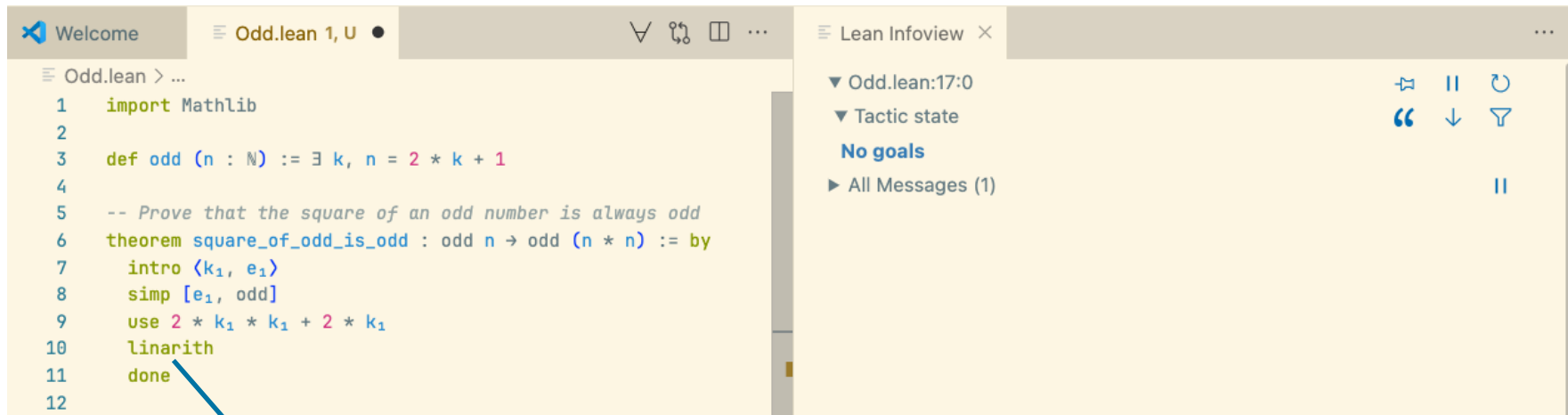
Machine-Checkable Proofs

You don't need to trust me to use my proofs.

You don't need to trust my automation to use it.

Code without fear.

Lean is a game where we can implement your own moves



The screenshot shows the Lean IDE interface. On the left, a code editor displays a proof script in `Odd.lean`. The script defines an `odd` function and proves a theorem about the square of an odd number. The `linarith` tactic is used to complete the proof. On the right, the 'Lean Infoview' panel shows the current tactic state, which is empty, indicating the proof is complete.

```

1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    intro ⟨k₁, e₁⟩
8    simp [e₁, odd]
9    use 2 * k₁ * k₁ + 2 * k₁
10   linarith
11   done
12

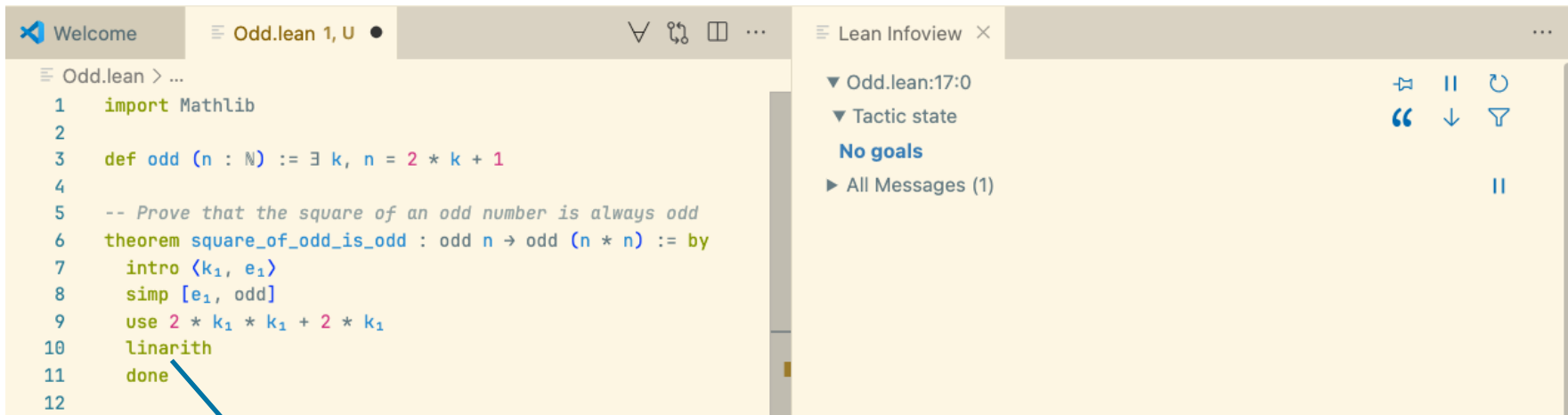
```

The right panel shows the tactic state:

- Odd.lean:17:0
- Tactic state
- No goals
- All Messages (1)

The `linarith` “move” was implemented by the Mathlib community in Lean!

Lean is a game where we can implement your own moves



The screenshot shows the Lean IDE interface. The left pane displays a Lean script in `Odd.lean` with the following content:

```

1  import Mathlib
2
3  def odd (n : ℕ) := ∃ k, n = 2 * k + 1
4
5  -- Prove that the square of an odd number is always odd
6  theorem square_of_odd_is_odd : odd n → odd (n * n) := by
7    intro ⟨k₁, e₁⟩
8    simp [e₁, odd]
9    use 2 * k₁ * k₁ + 2 * k₁
10   linarith
11   done
12

```

The right pane shows the `Lean Infoview` for the current goal, indicating "No goals" and "All Messages (1)".

The `linarith` “move” was implemented by the Mathlib community in Lean!

The `bv_decide` and `grind` “moves” are also implemented in Lean!



Lean FRO: Shaping the Future of Lean Development

The Lean Focused Research Organization (FRO) is a non-profit dedicated to Lean's development.

Founded in **August 2023**, the organization has 19 members.

Its mission is to enhance critical areas: **scalability, usability, documentation**, and **proof automation**.

It must reach **self-sustainability in August 2028** and become the **Lean Foundation**.

Philanthropic support is gratefully acknowledged from the **Simons Foundation**, the **Alfred P. Sloan Foundation**, **Richard Merkin**, and **Alex Gerko**.

Lean FRO: by numbers

20 releases and **4,383 pull requests** merged in the main repository only since its launch in July 2023.

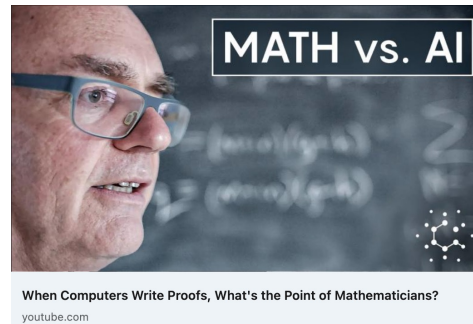
Public roadmaps: <https://lean-fro.org/about/roadmap-y2/>

Lean project was featured in multiple venues NY Times, Quanta, Scientific American, etc.



A.I. Is Coming for Mathematics, Too

For thousands of years, mathematicians have adapted to the latest advances in logic and reasoning. Are they ready for artificial intelligence?





How can I contribute?

Help building [Mathlib](#).

Want to engage with the vibrant Lean community? Join our [Zulip channel](#).

Interested in ML kernels? Contribute to the [KLR project](#).

Want to contribute to a large formalization project? Join the [FLT formalization project](#).

Start your own open-source Lean project! Your package will be available on our registry [Reservoir](#).

Start using Lean online: [live.lean-lang.org](#)

Support the Lean FRO: Funding, partnerships, or simply advocating the project.

Conclusion

Lean is an **efficient programming language** and **proof assistant**.

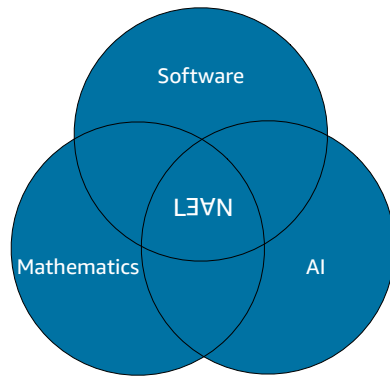
Lean is very extensible and is implemented in Lean.

Lean proofs are maintainable, stable, and transparent.

Progress is accelerating with the Lean FRO: module system, new compiler, new proof automation, etc.

The Mathlib community is changing how math is done.

It is not just about proving but also understanding complex objects and proofs, getting new insights, and navigating through the “thick jungles” that are **beyond our cognitive abilities**.



Thank You

<https://leanprover.zulipchat.com/>

x: @leanprover

LinkedIn: Lean FRO

Mastodon: @leanprover@functional.cafe

#leanlang, #leanprover

<https://www.lean-lang.org/>

